

Introduction to R

Advanced Statistics Lab

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Get R

From CRAN

RStudio

Ubuntu:

```
sudo apt install r-base
```

Some Introduction Material

From the developers: [Introduction to R](#)

RStudio: [Documentation](#)

[R for Data Science](#)

Background

- R is a general programming language with a specialization for statistics.
- R is generally used through a REPL and interpreted live.
- Functional programming paradigm: all about objects and functions.
- Major appeal: packages, graphics, and scientific workflows.
- Major drawback: scripts can grow without structure.

Assigning And Manipulating Variables

Note: R has two assignment operators, `<-` and `=`.

```
x = 5  
print(x)
```

```
[1] 5
```

```
x
```

```
[1] 5
```

```
y = x  
print(y)
```

```
[1] 5
```

```
x = 3  
  
print(x)
```

```
[1] 3
```

```
print(y)
```

```
[1] 5
```

Working With Data Frames

```
dat = data.frame(  
  name = c("Adam", "Paul", "Elly"),  
  age = c(22, 43, 54)  
)
```

```
str(dat)
```

```
'data.frame':   3 obs. of  2 variables:  
 $ name: chr  "Adam" "Paul" "Elly"  
 $ age : num  22 43 54
```

```
names = dat$name  
ages = dat[, "age"]
```

```
str(names)
```

```
chr [1:3] "Adam" "Paul" "Elly"
```

```
str(ages)
```

```
num [1:3] 22 43 54
```

Subsetting Data Frames

```
dat[dat$name == "Paul", ]
```

```
  name age  
2 Paul  43
```

```
dat[dat$age %in% c(22, 54), ]
```

```
  name age  
1 Adam  22  
3 Elly  54
```

```
dat$id = 1:nrow(dat)  
dat[, "id_2"] = dat$id + 2
```

```
print(dat)
```

```
  name age id id_2  
1 Adam  22  1    3  
2 Paul  43  2    4  
3 Elly  54  3    5
```

```
dat$id[dat$name == "Adam"] <- 12  
print(dat)
```

	name	age	id	id_2
1	Adam	22	12	3
2	Paul	43	2	4
3	Elly	54	3	5

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Reading And Writing Data

```
dat = data.frame(  
  name = c("Adam", "Paul", "Elly"),  
  age = c(22, 43, 54)  
)  
  
write.table(dat, file = "my_data.csv", row.names = FALSE, sep = ",")  
  
dat_read = read.table(file = "my_data.csv", sep = ",", header = TRUE)  
str(dat_read)
```

```
'data.frame':   3 obs. of  2 variables:  
 $ name: chr  "Adam" "Paul" "Elly"  
 $ age : int   22  43  54
```

Vectors And Matrices

```
x = as.integer(1)
str(x)
```

```
int 1
```

```
is.vector(x)
```

```
[1] TRUE
```

```
y = rnorm(10)
str(y)
```

```
num [1:10] -0.214 1.657 0.667 0.623 -1.155 ...
```

```
D = matrix(data = rnorm(5 * 5), nrow = 5, ncol = 5)
D
```

	[,1]	[,2]	[,3]	[,4]	[,5]
[1,]	-0.7921558	2.0568362	-1.41952524	-0.7012892	-0.81766062
[2,]	0.7289979	-0.3126469	-2.00600863	1.4872646	1.60369377
[3,]	-0.3210123	-0.7584070	-0.06071815	1.1154392	-0.07323955
[4,]	-1.4789812	0.7974553	1.09052207	-0.5539862	0.03396357
[5,]	0.4537841	1.5996937	0.89729598	1.4936323	1.21799820

```
class(D)
```

```
[1] "matrix" "array"
```

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Matrix Helpers

```
means = colMeans(D)
means
```

```
[1] -0.2818735  0.6765863 -0.2996868  0.5682121  0.3929511
```

```
d = diag(D)
d
```

```
[1] -0.79215578 -0.31264685 -0.06071815 -0.55398624  1.21799820
```

Data Exploration And Visualization

```
library(dlookr)
library(ggplot2)
```

```
data(iris)
str(iris)
```

```
'data.frame':  150 obs. of  5 variables:
 $ Sepal.Length: num  5.1 4.9 4.7 4.6 5 5.4 4.6 5 4.4 4.9 ...
 $ Sepal.Width : num  3.5 3 3.2 3.1 3.6 3.9 3.4 3.4 2.9 3.1 ...
 $ Petal.Length: num  1.4 1.4 1.3 1.5 1.4 1.7 1.4 1.5 1.4 1.5 ...
 $ Petal.Width : num  0.2 0.2 0.2 0.2 0.2 0.4 0.3 0.2 0.2 0.1 ...
 $ Species      : Factor w/ 3 levels "setosa","versicolor",...: 1 1 1 1 1 1 1 1 1 1
1 1 ...
```

```
dlookr::diagnose(iris)
```

```
# A tibble: 5 × 6
```

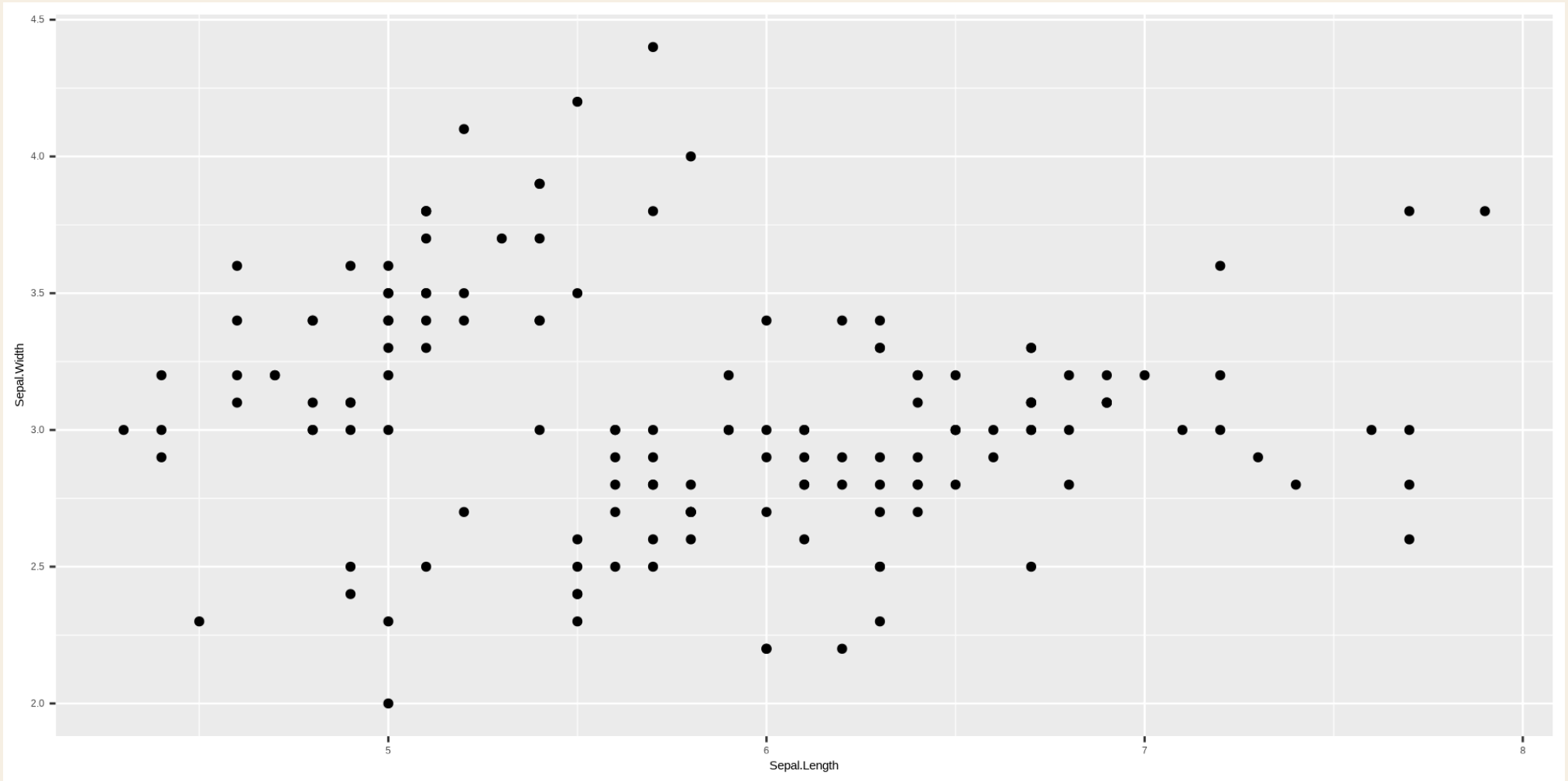
	variables	types	missing_count	missing_percent	unique_count	unique_rate
	<chr>	<chr>	<int>	<dbl>	<int>	<dbl>
1	Sepal.Length	numeric	0	0	35	0.233
2	Sepal.Width	numeric	0	0	23	0.153
3	Petal.Length	numeric	0	0	43	0.287

4	Petal.Width	numeric	0	0	22	0.147
5	Species	factor	0	0	3	0.02

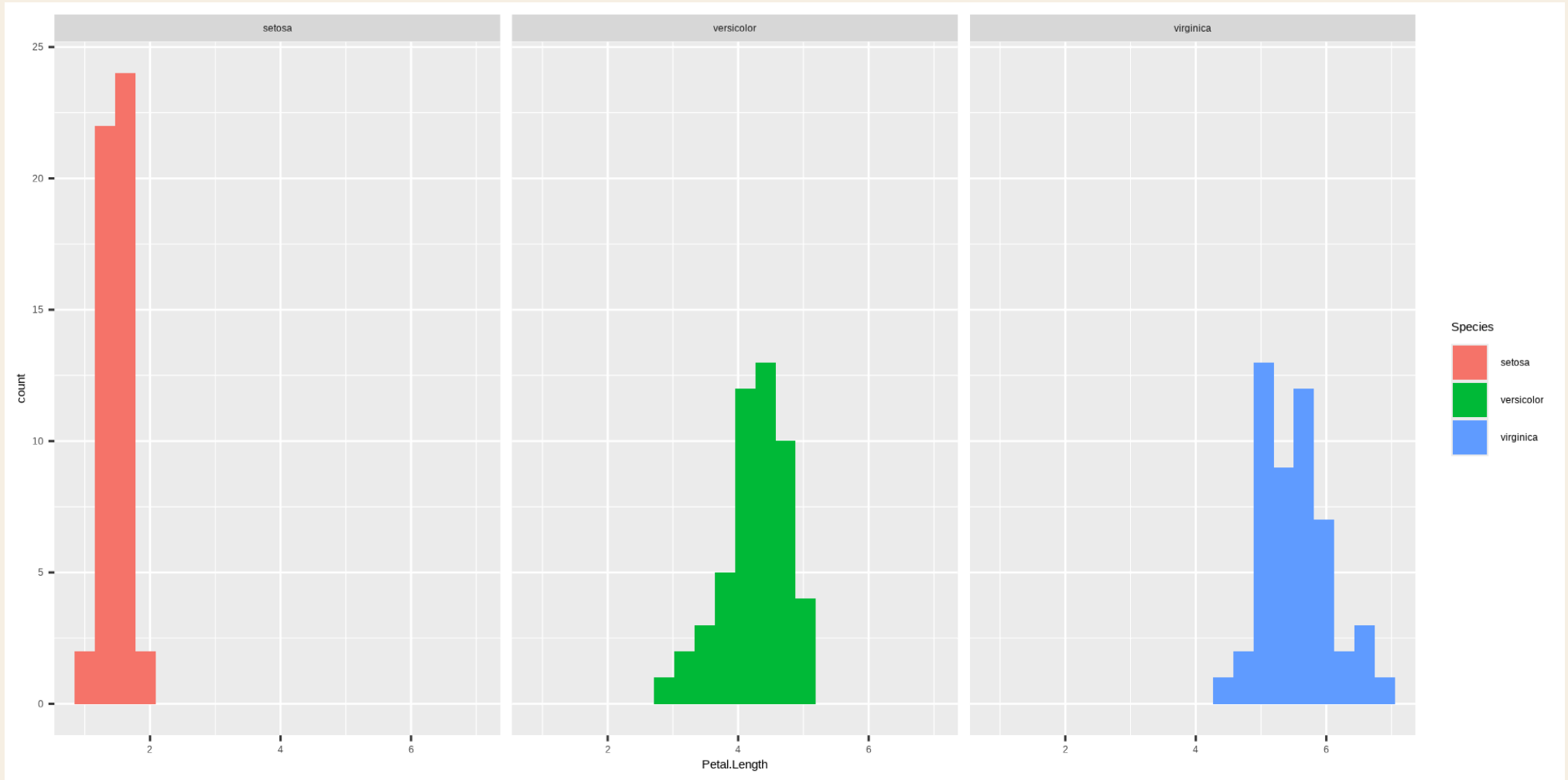
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Scatterplot

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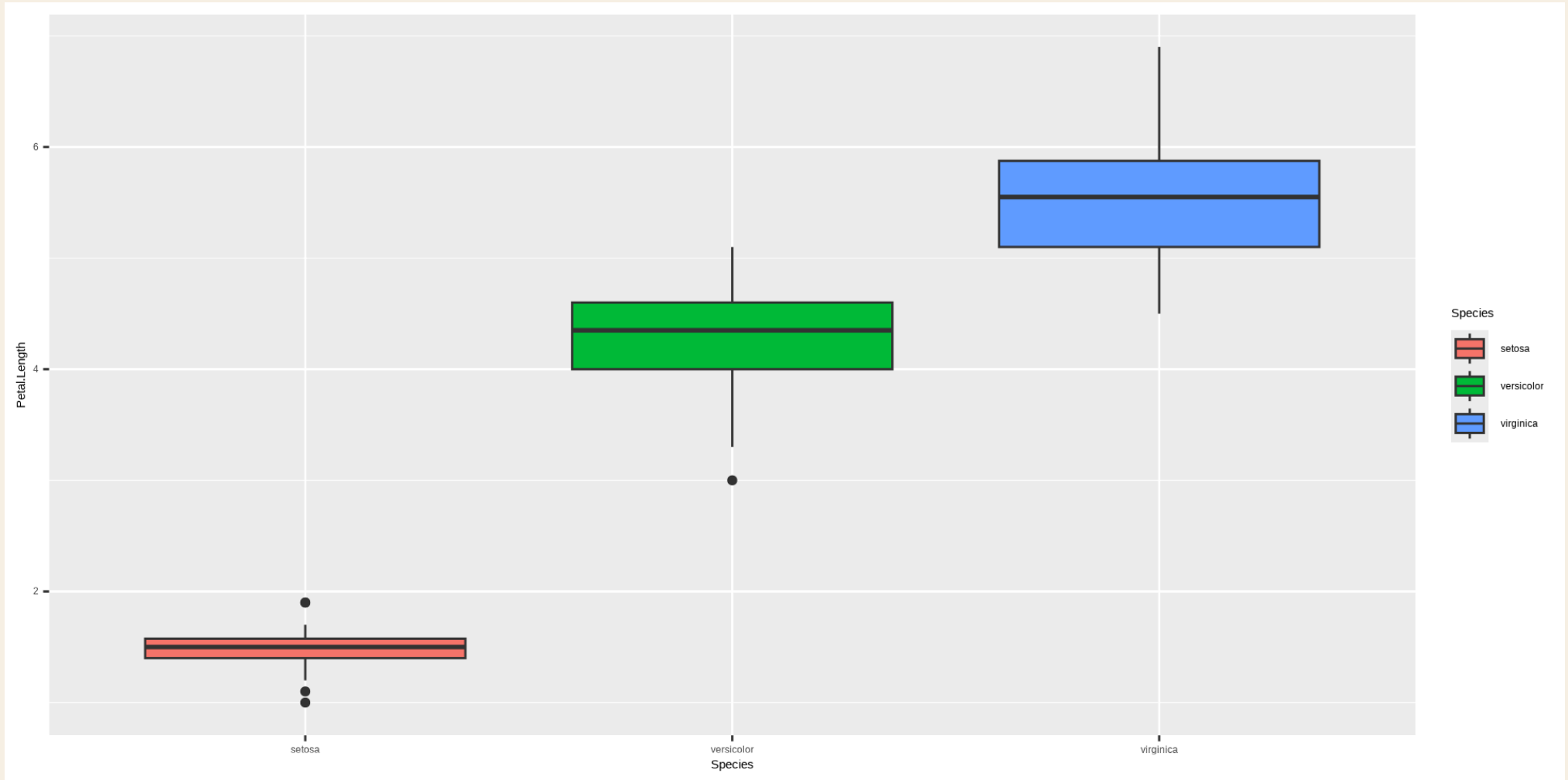


Histogram



Boxplot

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Writing Functions

All operators in R are functions.

```
?sum  
?`+`  
fix(`+`)
```

A Mean Function

```
my_mean_function = function(y) {  
  
  n = length(y)  
  sum = 0  
  for (i in 1:n) sum = sum + y[i]  
  
  mean = sum / n  
  
  return(mean)  
  
}
```

Use The Function

```
x = rnorm(1000, mean = 23.42, sd = 4)
```

```
our_mean = my_mean_function(x)  
our_mean
```

```
[1] 23.57816
```

```
mean(x)
```

```
[1] 23.57816
```